NEW YORK UNIVERSITY

Problem Set 1

Python Solutions

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Contents

1 Solving Linear Equations and Interpolation

1.1 Vandermonde system

We solve the linear system $Ax = b$ where $A \in \mathbb{R}^{n\times n}$ is a Vandermonde-type matrix with entries

$$A_{ij} = i^{n-j}, \qquad i, j \in \{1, \dots, n\},$$

and $b \in \mathbb{R}^n$ has entries

$$b_i = \sum_{k=0}^{n-1} i^k = \begin{cases} n, & i = 1, \\ \dfrac{i^n - 1}{i - 1}, & i > 1. \end{cases}$$

The true solution is $x = \mathbf{1}$. We report the conditioning of A and the numerical errors $\|x - \mathbf{1}\|_2$ and $\|Ax - b\|_2$ for $n \in \{5, 10, 15\}$.

Table 1: Vandermonde solve diagnostics

n	$\kappa_2(A)$	$\|x-\mathbf{1}\|_2$	$\|Ax-b\|_2$
5	2.62×10^4	3.20×10^{-13}	1.14×10^{-13}
10	2.11×10^{12}	7.89×10^{-7}	1.81×10^{-7}
15	2.58×10^{21}	2.17×10^2	2.29

where $\kappa_2(A)$ is the condition number of A in the 2-norm, which is defined as:

$$\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2, \tag{1}$$

where $\|A\|_2$ is the spectral norm of A, corresponding to the largest singular value of A. From Table 1, we see that the condition number grows rapidly as n increases, indicating that the Vandermonde system is ill-conditioned. Hence, for n values larger than 10, the solution is not accurate as columns become nearly linearly dependent in finite precision arithmetic.

1.2 Polynomial interpolation of $\log(s)$

We approximate $f(s) = \log(s)$ on $s \in [1, 10]$ by a polynomial of degree $n - 1$,

$$\hat{f}_n(s) = x_1 + x_2 s + \cdots + x_n s^{n-1}.$$

We fit coefficients by (overdetermined) least squares using $m = 100$ equally-spaced points. For $n \in \{5, 10, 15, 20\}$ we plot the approximant on a fine grid and the approximation error $\hat{f}_n(s) - \log(s)$, and we report the Euclidean norm of the error on the fine grid and the condition number of the regression matrix.

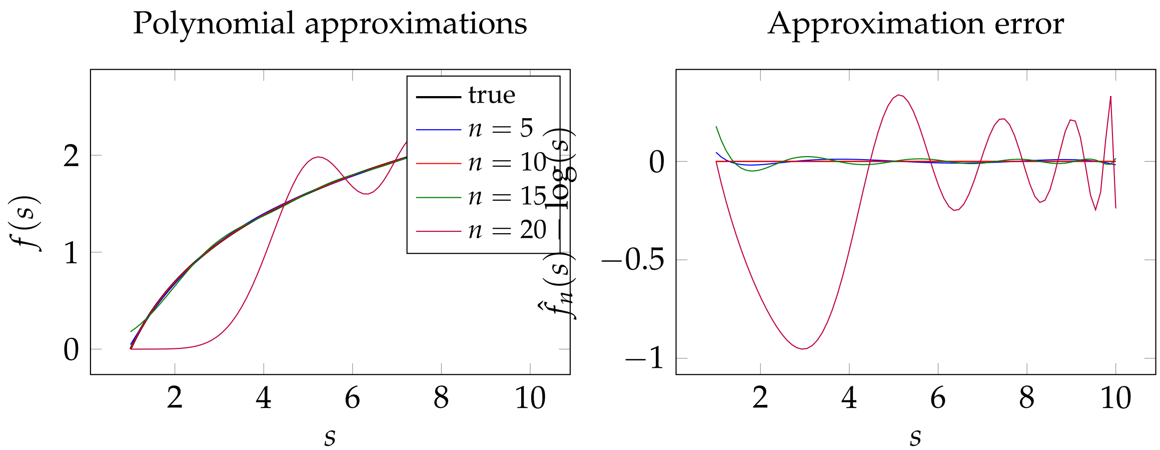


Figure 1: Polynomial approximations of $\log(s)$ (left) and approximation errors (right) for degrees $n - 1$ with $n \in \{5, 10, 15, 20\}$, using a monomial basis and least squares on $m = 100$ grid points.

The key numerical lesson from Problem 1.1 carries over directly to polynomial approximation: the least-squares design matrix has columns $1, s, s^2, \ldots, s^{n-1}$, so it is *Vandermonde-like* in the monomial basis. As n grows, the columns become increasingly ill-conditioned (similar to Table 1), so the fitted coefficients can become extremely sensitive to rounding error and to small perturbations in the data. Therefore, increasing n does not guarantee a uniformly better approximation: the function space is richer, but numerical instability can dominate and produce oscillatory or distorted fits (often most visible near the boundaries of $[1, 10]$) and non-monotone error behavior.

2 Optimal Labor Supply

We consider a static equilibrium with heterogeneous ability. There is a continuum of households indexed by ability $e \in [1, \bar{e}]$, distributed according to a truncated Pareto distribution $e \sim \widehat{F}(e)$. Each household chooses hours $h \geq 0$ to maximize CRRA utility separable in labor and consumption, subject to a budget constraint. Production is Cobb–Douglas in aggregate labor, with decreasing returns to scale, and the government levies a linear tax on labor income and rebates revenue lump-sum.

The goal is to compute the equilibrium allocation: a wage W, a lump-sum transfer T, and a schedule of hours $h(e)$ and consumption $c(e)$ such that (i) households optimize, (ii) firms maximize profits, and (iii) the goods market clears.

2.1 Setup

Ability is distributed with truncated Pareto cdf $\widehat{F}(e) = (1 - e^{-\alpha})/(1 - \bar{e}^{-\alpha})$ on $[1, \bar{e}]$. We discretize this distribution using Gauss–Legendre quadrature: nodes $\{e_i\}_{i=1}^n$ and weights $\{\omega_i\}_{i=1}^n$ with $\sum_i \omega_i = 1$, so that expectations are approximated by $\mathbb{E}[g(e)] \approx \sum_i \omega_i g(e_i)$. Aggregate labor supply is

$$L = \sum_{i=1}^{n} \omega_i e_i h_i. \tag{2}$$

Output is $Y = L^\eta$ with $\eta \in (0, 1)$. Under perfect competition, the wage is

$$W = \eta L^{\eta - 1}. \tag{3}$$

Tax revenue is rebated lump-sum; with tax rate τ on labor income, the transfer is

$$T = L^\eta - (1 - \tau) W L. \tag{4}$$

Household i has consumption

$$c_i = T + (1 - \tau) W e_i h_i. \tag{5}$$

With CRRA utility in consumption and disutility of labor, the first-order condition for interior $h_i > 0$ is

$$h_i^\gamma = (1 - \tau) W e_i c_i^{-\sigma}. \tag{6}$$

2.1.1 System of equations

Collect the vector of hours $\mathbf{h} = (h_1, \ldots, h_n)^\top$. Define aggregate objects as functions of $\mathbf{h}$:

$$\begin{aligned} L(\mathbf{h}) &= \sum_{i=1}^{n} \omega_i e_i h_i, \\ W(\mathbf{h}) &= \eta L(\mathbf{h})^{\eta-1}, \\ T(\mathbf{h}) &= L(\mathbf{h})^{\eta} - (1-\tau) W(\mathbf{h}) L(\mathbf{h}). \end{aligned} \tag{7}$$

Consumption of household i is $c_i(\mathbf{h}) = T(\mathbf{h}) + (1-\tau)W(\mathbf{h})e_i h_i$. The residual for the FOC of household i is

$$F_i(\mathbf{h}) = h_i^{\gamma} - (1-\tau)W(\mathbf{h}) e_i c_i(\mathbf{h})^{-\sigma}. \tag{8}$$

Equilibrium is a root of the n-dimensional system

$$\mathbf{F}(\mathbf{h}) = \mathbf{0}. \tag{9}$$

Goods market clearing is implied: $\sum_i \omega_i c_i = Y = L^{\eta}$ at a solution.

2.2 Solution methods

We implement two approaches: (1) solve $\mathbf{F}(\mathbf{h}) = \mathbf{0}$ in one shot with a robust Broyden method; (2) solve by fixed point on the wage W, with an inner Broyden step for the vector of hours given W.

2.2.1 Approach 1: Full system (Broyden on F)

We solve the coupled system $\mathbf{F}(\mathbf{h}) = \mathbf{0}$ directly. Because the Jacobian of $\mathbf{F}$ is dense and the system can be ill-scaled, we use the CompEcon-style Broyden method (inverse Jacobian approximation, backstepping line search, restarts), i.e. `broyden(..., method='compecon')`.

Algorithm 1 Full-system Broyden for labor supply equilibrium

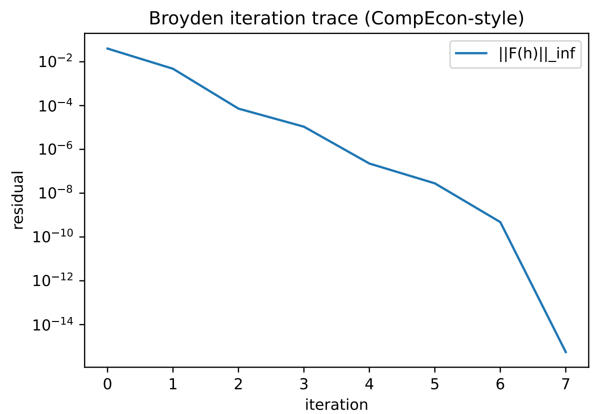
Input: Residual $\mathbf{F}$, initial guess $\mathbf{h}^{(0)}$, tolerance ε, max iterations K
Output: Equilibrium hours $\mathbf{h}^*$
$k \leftarrow 0$
while $k < K$ and $\|\mathbf{F}(\mathbf{h}^{(k)})\|_\infty > \varepsilon$ **do**
    Update $\mathbf{h}^{(k+1)}$ via Broyden (inverse Jacobian + line search)
    $k \leftarrow k + 1$
end while
return $\mathbf{h}^{(k)}$

2.2.2 Approach 2: Fixed point on wages (inner componentwise Broyden)

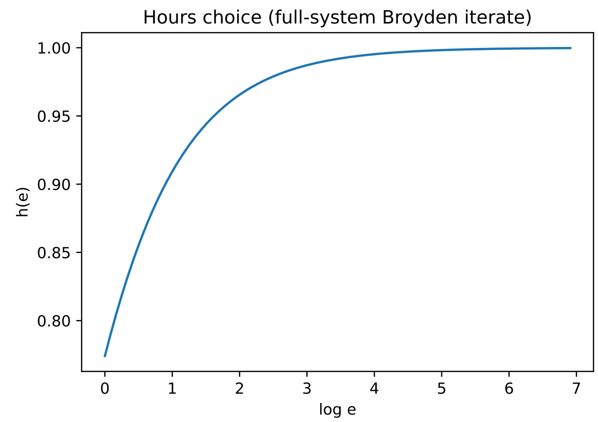
Given a guess for the wage W, the transfer is $T = L^\eta - (1-\tau)WL$ with $L = \sum_i \omega_i e_i h_i$. For fixed W and T, each household's FOC is a scalar equation in h_i with bounds $0 \leq h_i \leq \bar{h}$. We solve this vectorized system with safeguarded componentwise Broyden, i.e. `broyden(..., method='componentwise', a=0, b=`$\bar{h}$`)`. Then we aggregate L, update $W \leftarrow \eta L^{\eta-1}$ (optionally damped), and repeat until W converges.

Algorithm 2 Wage fixed-point with inner componentwise Broyden

Input: Parameters $(\tau, \eta, \sigma, \gamma)$, quadrature (e_i, ω_i), bounds $(\underline{h}, \bar{h})$, tolerance ε, max outer iterations K_{out}, max inner iterations K_{in}
Output: Equilibrium wage W^* and hours $\mathbf{h}^*$
$W \leftarrow W^{(0)}$ (e.g. initial guess)
$\mathbf{h} \leftarrow \mathbf{h}^{(0)}$ (e.g. $\mathbf{1}$)
for $k = 1$ to K_{out} **do**
    $L \leftarrow \sum_i \omega_i e_i h_i$; $T \leftarrow L^\eta - (1-\tau)WL$
    Solve for $\mathbf{h}$: FOC residuals in h_i given (W, T) equal zero, with $a_i = \underline{h}$, $b_i = \bar{h}$, using componentwise Broyden (max K_{in} iterations)
    $L_{\text{new}} \leftarrow \sum_i \omega_i e_i h_i$; $W_{\text{new}} \leftarrow \eta L_{\text{new}}^{\eta-1}$
    if $|W_{\text{new}} - W| < \varepsilon$ **then**
        return W_{new}, $\mathbf{h}$
    end if
    $W \leftarrow W + \lambda(W_{\text{new}} - W)$ (damping $\lambda \in (0, 1]$ optional)
end for
return W, $\mathbf{h}$



(a) Broyden iteration trace



(b) Equilibrium labor curve

Figure 2: Broyden convergence (Approach 1: full-system CompEcon-style) and equilibrium hours $h(e)$ vs. $\log e$.

2.3 Implementation notes

- **Quadrature (Sec. 2.1):** We use Gauss–Legendre nodes and weights on $[1, \bar{e}]$ and multiply weights by the truncated Pareto pdf so that $\sum_i \omega_i = 1$. We check that the first three moments match the analytical formulas.
- **Approach 1:** We call `broyden(..., method='compecon')` with guardrails in the residual (clip h to $[10^{-12}, \bar{h}]$, penalize $c \leq 0$) to avoid numerical explosions.
- **Approach 2:** We call `broyden(..., method='componentwise', a=hmin, b=hmax)` for the inner FOC system given (W, T). The outer loop updates W until $|W_{\text{new}} - W| < \varepsilon$.
- **Checks:** We verify goods market clearing $\sum_i \omega_i c_i = L^\eta$ and report the FOC residual norm and transfer T, labor L, output Y, and wage W.

3 Calibration by Simulation

We consider an economy with a continuum of firms indexed by i. Each firm i has idiosyncratic productivity z_{it} and chooses labor demand $\ell_{it} \geq 0$. Output is produced with a Cobb-Douglas technology in productivity and labor:

$$y_{it} = (z_{it})^{1-\eta} (\ell_{it})^{\eta}, \tag{10}$$

where $\eta \in (0, 1)$. In a partial equilibrium setting we normalize the wage $W = 1$. The firm's problem is

$$\max_{\ell_{it} \geq 0} \quad y_{it} - W\ell_{it} = (z_{it})^{1-\eta} (\ell_{it})^{\eta} - \ell_{it}. \tag{11}$$

The first-order condition $\eta (z_{it})^{1-\eta} (\ell_{it})^{\eta-1} = 1$ yields optimal labor

$$\ell_{it} = \eta^{1/(1-\eta)} z_{it}, \tag{12}$$

and therefore output

$$y_{it} = (z_{it})^{1-\eta} \left(\eta^{1/(1-\eta)} z_{it}\right)^{\eta} = \eta^{\eta/(1-\eta)} z_{it}. \tag{13}$$

Hence $\log y_{it} = \log z_{it} + \text{const}$: (log) output and (log) productivity coincide up to a constant. We observe a panel of log productivity (or log output) and seek to calibrate the parameters of the stochastic process that governs $\log z_{it}$ by matching selected moments.

3.1 Stochastic process for log productivity

We drop the firm index i when describing the law of motion. Let $y_t \equiv \log z_{it}$ denote log productivity (or log output) and $x_t \equiv \log \hat{z}_{it}$ the persistent component. The process is

$$\begin{aligned}
&y_t = x_t + \varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2), \\
&x_t = \rho x_{t-1} + u_t, \qquad u_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_u^2), \qquad |\rho| < 1, \\
&u_t \perp \varepsilon_s \quad \forall t, s.
\end{aligned}$$

So log productivity is the sum of a persistent AR(1) component x_t and an iid transitory shock ε_t.

3.2 Analytical moments

3.2.1 Variance and standard deviation of $\log y_t$

The persistent component has stationary variance $\mathrm{Var}(x_t) = \sigma_u^2/(1-\rho^2)$. Hence

$$\begin{aligned}
\mathrm{Var}(y_t) &= \mathrm{Var}(x_t) + \mathrm{Var}(\varepsilon_t) = \frac{\sigma_u^2}{1-\rho^2} + \sigma_\varepsilon^2, \\
\mathrm{sd}(y_t) &= \sqrt{\frac{\sigma_u^2}{1-\rho^2} + \sigma_\varepsilon^2}.
\end{aligned}$$

3.2.2 Variance and standard deviation of $\Delta y_t \equiv y_t - y_{t-1}$

Using $\mathrm{Var}(\Delta y_t) = 2\,\mathrm{Var}(y_t) - 2\,\mathrm{Cov}(y_t, y_{t-1})$ and $\mathrm{Cov}(y_t, y_{t-1}) = \mathrm{Cov}(x_t, x_{t-1}) = \rho\,\mathrm{Var}(x_t) = \rho\sigma_u^2/(1-\rho^2)$,

$$\begin{aligned}
\mathrm{Var}(\Delta y_t) &= 2\left(\frac{\sigma_u^2}{1-\rho^2} + \sigma_\varepsilon^2\right) - 2\,\rho\,\frac{\sigma_u^2}{1-\rho^2} = 2\left(\frac{\sigma_u^2}{1+\rho} + \sigma_\varepsilon^2\right), \\
\mathrm{sd}(\Delta y_t) &= \sqrt{2\left(\frac{\sigma_u^2}{1+\rho} + \sigma_\varepsilon^2\right)}.
\end{aligned}$$

3.2.3 Autocorrelations of $\log y_t$

For $k \geq 1$,

$$\begin{aligned}
\mathrm{Cov}(y_t, y_{t-k}) &= \mathrm{Cov}(x_t, x_{t-k}) = \rho^k\,\mathrm{Var}(x_t) = \rho^k\,\frac{\sigma_u^2}{1-\rho^2}, \\
\mathrm{Corr}(y_t, y_{t-k}) &= \frac{\mathrm{Cov}(y_t, y_{t-k})}{\mathrm{Var}(y_t)} = \rho^k \cdot \frac{\frac{\sigma_u^2}{1-\rho^2}}{\frac{\sigma_u^2}{1-\rho^2} + \sigma_\varepsilon^2}.
\end{aligned}$$

We target the autocorrelations at lags $k = 1$, 3, and 5:

$$\mathrm{Corr}(y_t, y_{t-1}) = \rho \cdot \frac{\frac{\sigma_u^2}{1-\rho^2}}{\frac{\sigma_u^2}{1-\rho^2} + \sigma_\varepsilon^2},$$

$$\mathrm{Corr}(y_t, y_{t-3}) = \rho^3 \cdot \frac{\frac{\sigma_u^2}{1-\rho^2}}{\frac{\sigma_u^2}{1-\rho^2} + \sigma_\varepsilon^2},$$

$$\mathrm{Corr}(y_t, y_{t-5}) = \rho^5 \cdot \frac{\frac{\sigma_u^2}{1-\rho^2}}{\frac{\sigma_u^2}{1-\rho^2} + \sigma_\varepsilon^2}.$$

3.3 Simulation and target moments

We simulate a panel of $N = 100{,}000$ firms over $T = 10$ periods. For each firm we draw the initial persistent component from the ergodic distribution $\mathcal{N}(0, \sigma_u^2/(1-\rho^2))$, then evolve x_t and add iid ε_t to obtain $y_t = x_t + \varepsilon_t$. From the simulated panel we compute:

- standard deviation of $\log y_t$,
- standard deviation of the one-year change $\Delta y_t = y_t - y_{t-1}$,
- autocorrelations of y_t at lags 1, 3, and 5.

We match these to the following target moments (from the data): $\mathrm{sd}(y) = 1.31$, $\mathrm{sd}(\Delta y) = 0.59$, $\mathrm{Corr}(y_t, y_{t-1}) = 0.90$, $\mathrm{Corr}(y_t, y_{t-3}) = 0.87$, $\mathrm{Corr}(y_t, y_{t-5}) = 0.85$.

Figure 3 shows the distribution of log-productivity changes at horizons 1, 3, and 5 years from the simulated panel (using initial parameter guesses). Under this two-component process, the change $\Delta_k y_t = y_t - y_{t-k}$ is the sum of two independent normal random variables and is therefore normal; the empirical densities align with this.

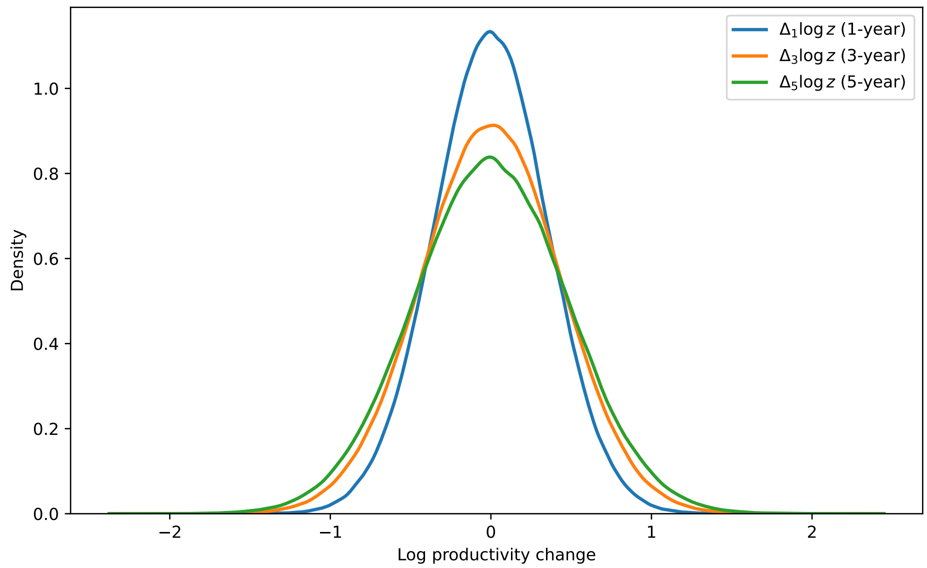


Figure 3: Distribution of log-productivity changes: density of $\Delta_1 \log z$, $\Delta_3 \log z$, and $\Delta_5 \log z$ from the simulated panel (KDE).

3.4 Estimation: Nelder-Mead

We estimate $\theta = (\rho, \sigma_u, \sigma_\varepsilon)$ by minimizing the sum of squared errors between the five model moments and the five target moments (SSE). We use the Nelder-Mead simplex algorithm (no derivatives), with simple bounds: $\rho \in [0.70, 0.99]$, $\sigma_u, \sigma_\varepsilon \in [0.01, 0.50]$. At each iteration we simulate the panel once (fixed seed for reproducibility) and compute the moments; the objective is the squared Euclidean norm of the moment residual vector.

Figure 4 plots the SSE at each Nelder-Mead iteration. The algorithm converges to a low SSE, and the resulting parameter vector reproduces the target moments closely (e.g. $\hat{\rho} \approx 0.986$, $\hat{\sigma}_u \approx 0.21$, $\hat{\sigma}_\varepsilon \approx 0.39$).

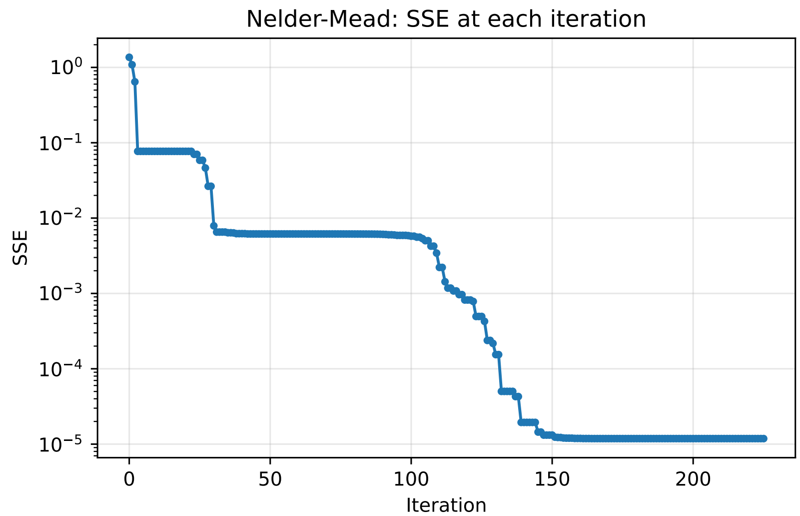


Figure 4: Nelder-Mead: SSE (sum of squared moment errors) at each iteration.

3.5 Implementation notes

- **Simulation:** Initial x_0 is drawn from $\mathcal{N}(0, \sigma_u^2/(1-\rho^2))$; then $x_t = \rho x_{t-1} + u_t$ and $y_t = x_t + \varepsilon_t$ for $t = 1, \dots, T-1$.

- **Moments:** Standard deviations and correlations are computed from the flattened panel (all firms and time periods used for each moment).

- **Nelder-Mead:** We use `hetmacro.optimize.nelder_mead` with `return_info=True` to record the objective value at each iteration for the convergence plot.

3.6 Extension: Jump-diffusion (fat tails)

To capture fat tails and leptokurtic distributions in the data, we extend the persistent component to a *jump-diffusion*. Idiosyncratic log productivity is $\tilde{y}_{it} = x_{it} + \varepsilon_{it}$ with $\varepsilon_{it} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. The persistent state follows

$$
\begin{aligned}
x_{it} &= \rho x_{i,t-1} + \eta_{it} + J_{it}, \qquad \eta_{it} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \\
J_{it} &= b_{it}\,\kappa_{it}, \qquad b_{it} \sim \text{Bernoulli}(\pi), \quad \kappa_{it} \sim \mathcal{N}(\mu_J, \sigma_J^2),
\end{aligned}
$$

with $\varepsilon_{it}, \eta_{it}, b_{it}, \kappa_{it}$ mutually independent. The jump term delivers occasional large moves in $\Delta\tilde{y}_{it}$, generating fat tails while preserving the AR(1)+noise structure. The stationary variance of x is $\text{Var}(x) = (\sigma_\eta^2 + \text{Var}(J))/(1-\rho^2)$ with $\text{Var}(J) = \pi\sigma_J^2 + \pi(1-\pi)\mu_J^2$; we draw x_0 from $\mathcal{N}(0, \sqrt{\text{Var}(x)})$.

We estimate $\theta = (\rho, \sigma_\eta, \sigma_\varepsilon, \pi, \mu_J, \sigma_J)$ by minimizing the same SSE over the same five target moments (simulation-based; no closed-form moments for the jump model). Bounds: $\rho \in [0.70, 0.99]$, $\sigma_\eta, \sigma_\varepsilon \in [0.01, 0.50]$, $\pi \in [0.01, 0.5]$, $\mu_J \in [-1, 1]$, $\sigma_J \in [0.01, 1]$. Figure 5 shows the Nelder-Mead SSE trace for the 6-parameter problem; Figure 6 shows the distribution of log-productivity changes under the jump-diffusion fit, with fatter tails than the baseline.

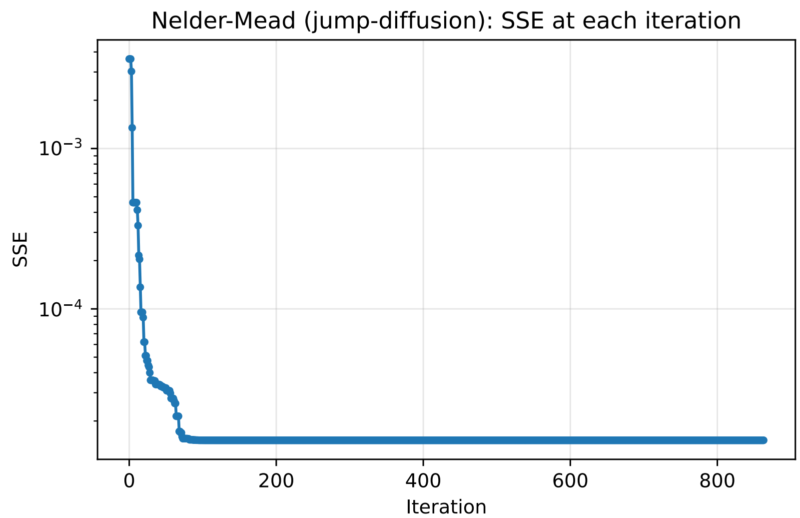


Figure 5: Nelder-Mead (jump-diffusion): SSE at each iteration.

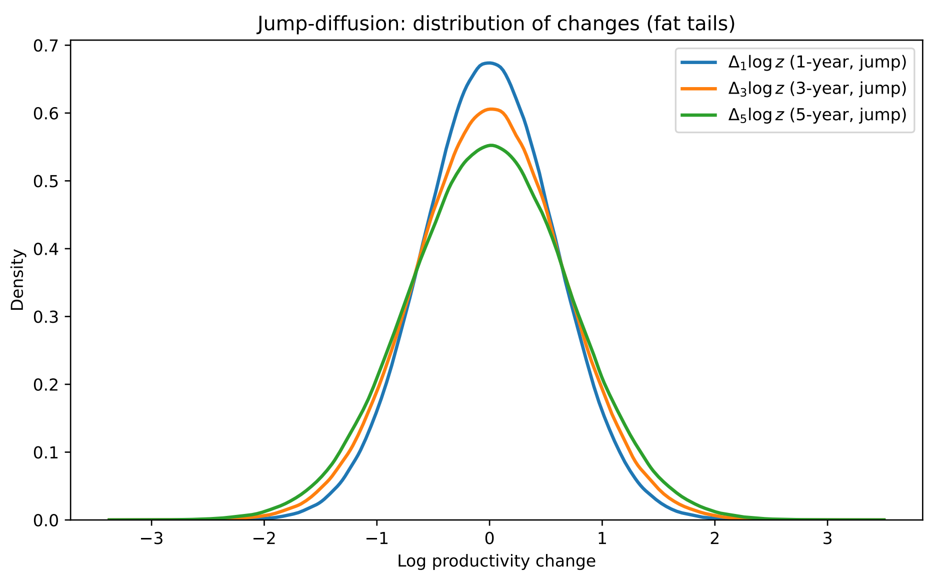


Figure 6: Jump-diffusion: distribution of log-productivity changes (fat tails).

Table 2 reports target vs. model moments for both specifications; Table 3 reports the calibrated parameters. Tables are generated by `gen_problem3_figures.py`.

Table 2: Target vs. model moments

Moment	Target	Baseline (AR(1)+noise)	Jump-diffusion
$\mathrm{sd}(\log y)$	1.310	1.310	1.310
$\mathrm{sd}(\Delta \log y)$	0.590	0.590	0.590
$\mathrm{Corr}(y_t, y_{t-1})$	0.900	0.899	0.899
$\mathrm{Corr}(y_t, y_{t-3})$	0.870	0.873	0.873
$\mathrm{Corr}(y_t, y_{t-5})$	0.850	0.849	0.848

Table 3: Calibrated parameters

Parameter	Model	Estimate
ρ	Baseline	0.986
σ_u	Baseline	0.210
σ_ε	Baseline	0.390
ρ	Jump-diffusion	0.986
σ_η	Jump-diffusion	0.198
σ_ε	Jump-diffusion	0.390
π	Jump-diffusion	0.050
μ_J	Jump-diffusion	0.000
σ_J	Jump-diffusion	0.314

4 Two-Period Labor Supply

A consumer lives for two periods with preferences

$$V = \sum_{t=1}^{2} \beta^{t-1} \left(\frac{c_t^{1-\sigma}}{1-\sigma} - \frac{h_t^{1+\gamma}}{1+\gamma} \right), \tag{14}$$

where σ is the inverse elasticity of intertemporal substitution and γ is the inverse Frisch elasticity of labor supply. The consumer saves in a riskless asset with interest rate r and receives income $T + (1-\tau)We_t h_t$ in period t. The budget constraints are

$$c_1 + a_2 = T + (1-\tau)We_1 h_1, \tag{15}$$

$$c_2 = T + (1-\tau)We_2 h_2 + (1+r)a_2. \tag{16}$$

Efficiency is normalized to $e_1 = 1$ in the first period and equals e_2 in the second; we solve for different values of e_2. The consumer cannot borrow:

$$a_2 \geq 0. \tag{17}$$

Parameters. We use $\beta = 0.90$, $\sigma = 1.5$, $\gamma = 2$, $r = 0.04$, $W = 1$, $\tau = 0.25$, $T = 0.15$.

4.1 Backward induction and optimality conditions

The problem is solved by backward induction, respecting the no-borrowing constraint. In period 2, the consumer enters with assets a and chooses (c_2, h_2) subject to the budget constraint and the labor-leisure first-order condition. In period 1, the consumer chooses (c_1, h_1) (and hence a_2) to maximize lifetime utility, subject to $a_2 \geq 0$.

Period 2. The labor-leisure FOC is

$$h_2^{\gamma} = (1-\tau)We_2\, c_2^{-\sigma}. \tag{18}$$

With $c_2 = T + (1-\tau)We_2 h_2 + (1+r)a$, substituting the budget into the FOC yields a single equation in c_2. Equivalently, define next-period assets as a function of current consumption and (implied) hours:

$$a' = T + (1-\tau)Weh + (1+r)a - c. \tag{19}$$

Given c, hours are backed out from the FOC: $h = \left((1-\tau)We\,c^{-\sigma}\right)^{1/\gamma}$. At the end of life we require $a_3 = 0$, so we solve for c_2 such that $a'(c_2) = 0$. This yields $c_2(a, e_2)$, $h_2(a, e_2)$, and the period-2 value

$$V_2(a, e_2) = \frac{c_2(a, e_2)^{1-\sigma}}{1-\sigma} - \frac{h_2(a, e_2)^{1+\gamma}}{1+\gamma}. \tag{20}$$

Period 1. For any c_1, hours satisfy $h_1^{\gamma} = (1-\tau)We_1 c_1^{-\sigma}$ (with $e_1 = 1$) and savings are

$$a_2 = T + (1-\tau)We_1 h_1 - c_1. \tag{21}$$

The upper bound $\bar{c}_1$ such that $a_2(\bar{c}_1) = 0$ ensures the no-borrowing constraint when we restrict $c_1 \in (0, \bar{c}_1]$. The consumer solves

$$\max_{c_1 \in (0, \bar{c}_1]} \quad \frac{c_1^{1-\sigma}}{1-\sigma} - \frac{h_1^{1+\gamma}}{1+\gamma} + \beta V_2(a_2, e_2). \tag{22}$$

At an interior optimum the Euler equation holds: $c_1^{-\sigma} = \beta(1+r)c_2^{-\sigma}$.

4.2 Numerical solution: period 2

We implement the period-2 solution as follows. For each pair (a, e_2) on a grid, we find c_2 such that $a'(c_2) = 0$ using a bisection root-finder on the function $\texttt{savings}(c, a, e_2, p)$, which returns a' given consumption (and implied hours from the labor FOC). This yields $c_2(a, e_2)$, $h_2(a, e_2)$, and $V_2(a, e_2)$.

Algorithm 3 Period-2 solution (for each (a, e_2))

1: **Input:** Asset grid a, productivity e_2, parameters p
2: **Output:** $c_2(a, e_2)$, $h_2(a, e_2)$, $V_2(a, e_2)$
3: **for** each (a_i, e_2) **do**
4: Bracket $c \in (0, \overline{\text{coh}})$ so that $\texttt{savings}(c, a_i, e_2, p)$ has a root
5: $c_2 \leftarrow \texttt{bisect}(\texttt{savings}, \ell, u; a_i, e_2, p)$ so that $a' = 0$
6: $h_2 \leftarrow$ hours from labor FOC given c_2
7: $V_2 \leftarrow u(c_2, h_2)$
8: **end for**
9: **return** V_2, c_2, h_2

Figure 7 shows the period-2 consumption, hours, and value as surfaces over (a, e_2).

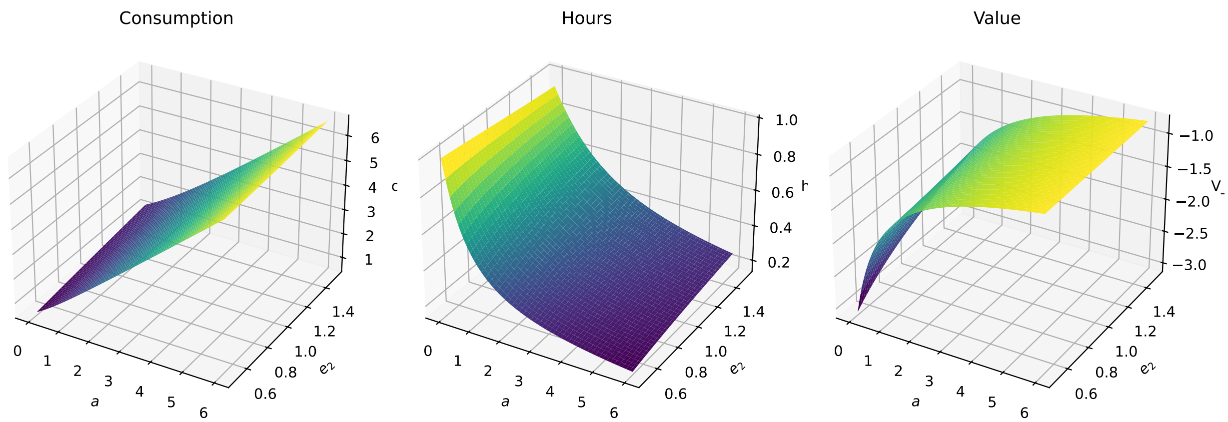


Figure 7: Period-2 policy and value: $c_2(a, e_2)$, $h_2(a, e_2)$, and $V_2(a, e_2)$ on a grid of assets a and second-period productivity e_2.

4.3 Numerical solution: period 1

Given the period-2 value and policies, we solve the period-1 problem for each e_2 in a grid. Step (A): find the maximum feasible period-1 consumption $\bar{c}_1$ such that $a_2 = 0$ (bisection on $\texttt{savings}(c, 0, 1, p)$ with $a_1 = 0$, $e_1 = 1$). Step (B): maximize the period-1 lifetime value over $c_1 \in (0, \bar{c}_1]$ using golden-section search on the function that returns $u(c_1, h_1) + \beta V_2(a_2, e_2)$. Step (C): from the optimal c_1 we recover h_1, a_2, and then c_2, h_2 from the period-2 solution at (a_2, e_2).

Algorithm 4 Period-1 solution (for each e_2)

Input: Productivity grid e_2, parameters p
Output: $c_1(e_2)$, $h_1(e_2)$, $a_2(e_2)$, $c_2(e_2)$, $h_2(e_2)$
for each e_2 **do**
  $\bar{c}_1 \leftarrow \texttt{bisect}(\texttt{savings}, \ell, u; 0, 1, p)$ so that $a_2 = 0$
  $c_1^* \leftarrow \arg\max_{c_1 \in (0, \bar{c}_1]} \{u(c_1, h_1) + \beta V_2(a_2, e_2)\}$ via `golden_search` on minus the objective
  $h_1 \leftarrow$ hours from labor FOC; $a_2 \leftarrow \texttt{savings}(c_1^*, 0, 1, p)$
  $(c_2, h_2) \leftarrow$ period-2 solution at (a_2, e_2)
end for
return c_1, h_1, a_2, c_2, h_2

The policy functions we report are expressed as functions of e_2 only. This is because we condition on the asset choice a_2 being the one chosen optimally in period 1 given e_2 (and $a_1 = 0$). So for each e_2, the plotted c_1, h_1, a_2, c_2, h_2 are the equilibrium choices along that path; they are not the full policy functions $c_2(a, e_2)$ and $h_2(a, e_2)$ over all a, but the realized consumption and hours given the endogenous $a_2(e_2)$.

We plot consumption, hours, and assets only for *interior* points where $a_2 > 0$ (non-binding borrowing constraint), so that the Euler equation is approximately satisfied and the curves are smooth.

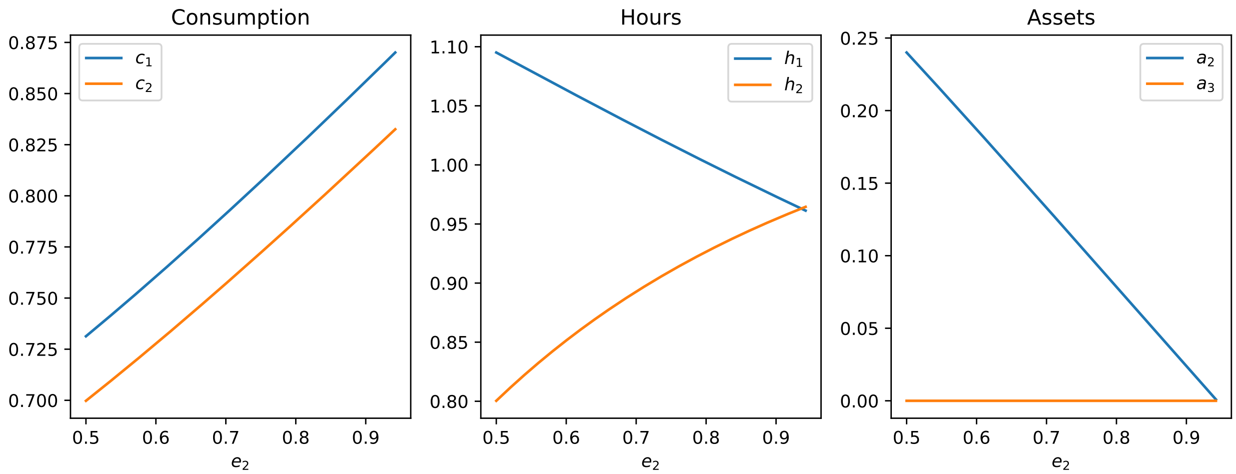


Figure 8: Period-1 policy functions as functions of e_2: consumption (c_1, c_2), hours (h_1, h_2), and assets (a_2, a_3). Plotted only where $a_2 > 0$ (interior). The second-period choices are evaluated at the endogenously chosen $a_2(e_2)$.

5 Problem 5: Two-Period Portfolio Choice

The consumer lives for two periods and has Epstein–Zin preferences

$$V(m) = \left[c_1^{1-\gamma} + \beta \left(\mathbb{E}\, c_2^{1-\sigma} \right)^{\frac{1-\gamma}{1-\sigma}} \right]^{\frac{1}{1-\gamma}}, \tag{23}$$

where γ is inverse EIS and σ is risk aversion. The consumer begins period 1 with cash-on-hand $m > 0$ and chooses (i) the savings share $x \geq 0$ and (ii) the risky share of savings $x_s \in [0, 1]$ (no borrowing, no short-selling). Period-1 consumption is

$$c_1 = (1 - x)m. \tag{24}$$

The consumer can invest savings xm in a safe asset with gross return $R_f = 1 + r$ or a risky asset with gross return $R = 1 + r + \varepsilon_2$. Period-2 consumption is

$$c_2 = y_2 + (1 + r)xm + \varepsilon_2 x_s xm. \tag{25}$$

The pair $(\log y_2, \varepsilon_2)$ is jointly normal:

$$\begin{bmatrix} \log y_2 \\ \varepsilon_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ \mu \end{bmatrix}, \begin{bmatrix} \sigma_y^2 & \rho\sigma_y\sigma_\varepsilon \\ \rho\sigma_y\sigma_\varepsilon & \sigma_\varepsilon^2 \end{bmatrix} \right). \tag{26}$$

We solve the problem for two correlations, $\rho \in \{0.3, 0.6\}$.

Parameters. We use $\beta = 0.90$, $\sigma = 5$, $\gamma = 1/2$, $r = 0.04$, $\mu = 0.06$, $\sigma_\varepsilon = 0.15$, and $\sigma_y = 0.20$. We solve for a grid of cash-on-hand values $m \in [0.1, 10]$.

5.1 Optimality conditions and Euler equations

Define the period-2 certainty equivalent

$$CE_2 \equiv \left(\mathbb{E}\, c_2^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \tag{27}$$

Since the outer power in (23) is monotone, the choice problem is equivalent to maximizing

$$J \equiv c_1^{1-\gamma} + \beta CE_2^{1-\gamma}. \tag{28}$$

Let the stochastic discount factor associated with these preferences be

$$M_2 \equiv \left(\frac{c_2}{c_1}\right)^{-\gamma}\left(\frac{c_2}{CE_2}\right)^{\gamma-\sigma}. \tag{29}$$

At an interior optimum, the Euler equations for the safe and risky assets are

$$\begin{aligned} 1 &= \beta\,\mathbb{E}\left[M_2(1+r)\right], \\ 1 &= \beta\,\mathbb{E}\left[M_2(1+r+\varepsilon_2)\right]. \end{aligned} \tag{30}$$

These are the equations we use for diagnostic checks. When the constraints bind (e.g. $x = 0$ or $x_s \in \{0, 1\}$), the corresponding Euler equation need not hold with equality.

5.2 Numerical solution

5.2.1 Quadrature for $(\log y_2, \varepsilon_2)$

We approximate expectations with tensor-product Gauss–Hermite quadrature. This choice is natural because $(\log y_2, \varepsilon_2)$ is bivariate normal: Gauss–Hermite quadrature is designed for integrals with a Gaussian (normal) weight function, and the multivariate version uses a tensor product of one-dimensional Gauss–Hermite rules, matching the joint normal density.

The integrand $g(y_2, \varepsilon_2)$ in the quadrature is the object whose expectation we need. For the Euler equations (30) this is the stochastic discount factor times the asset's gross return: $g = M_2 \times$ (gross return). The SDF M_2 depends on (y_2, ε_2) through period-2 consumption $c_2 = y_2 + (1+r)xm + \varepsilon_2 x_s xm$. Concretely, for the safe-asset equation we use $g_{\text{safe}}(y_2, \varepsilon_2) = M_2(y_2, \varepsilon_2)\,(1+r)$; for the risky-asset equation we use $g_{\text{risky}}(y_2, \varepsilon_2) = M_2(y_2, \varepsilon_2)\,(1+r+\varepsilon_2)$. With nodes $\{(\log y_2^j, \varepsilon_2^j)\}_{j=1}^J$ and weights $\{w_j\}_{j=1}^J$ (with $\sum_j w_j = 1$), we set $y_2^j = \exp(\log y_2^j)$ and approximate

$$\mathbb{E}[g(y_2, \varepsilon_2)] \approx \sum_{j=1}^{J} w_j\, g(y_2^j, \varepsilon_2^j). \tag{31}$$

We use 7 nodes in each dimension (so $J = 49$).

5.2.2 Optimization over (x, x_s)

For each m and each ρ, we solve the bounded two-dimensional maximization problem

$$\max_{x \in [0,1],\, x_s \in [0,1]} V(m; x, x_s), \tag{32}$$

where V is defined by (23)–(25). We solve via a bounded quasi-Newton optimizer (L-BFGS-B) on $-V$, using `hetmacro.optimize.minimize_lbfgsb`.

Algorithm 5 Solve policies for each m and ρ

Input: m grid, parameters p, correlation ρ
Output: policy functions $x(m)$ and $x_s(m)$
Build quadrature nodes/weights for $(\log y_2, \varepsilon_2)$
for each m **do**
    Solve $\max_{x, x_s \in [0,1]} V(m; x, x_s)$ via L-BFGS-B
end for
return $x(m)$, $x_s(m)$

5.3 Results and Euler diagnostics

Figure 9 plots the optimal saving share $x(m)$ and risky share $x_s(m)$ for $\rho = 0.3$ and $\rho = 0.6$. Figure 10 shows Euler residuals for the safe and risky asset, computed from (30) and masked to points where both controls are interior (away from the bounds).

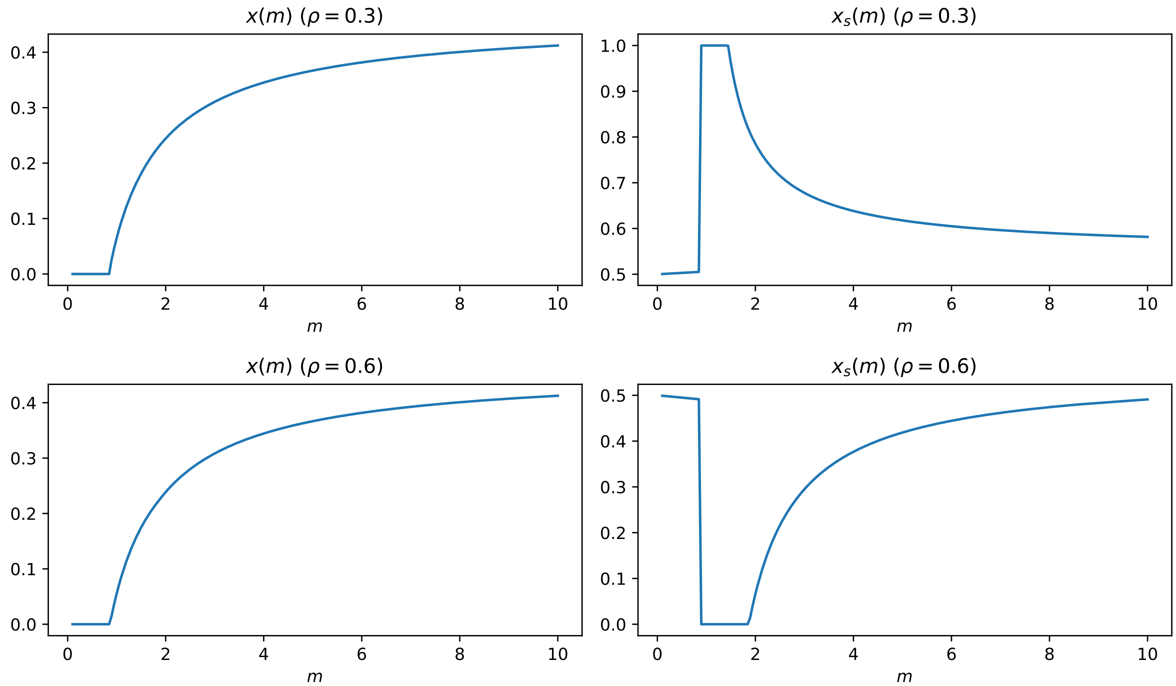


Figure 9: Problem 5 policy functions: saving share $x(m)$ and risky share $x_s(m)$ for correlations $\rho = 0.3$ and $\rho = 0.6$.

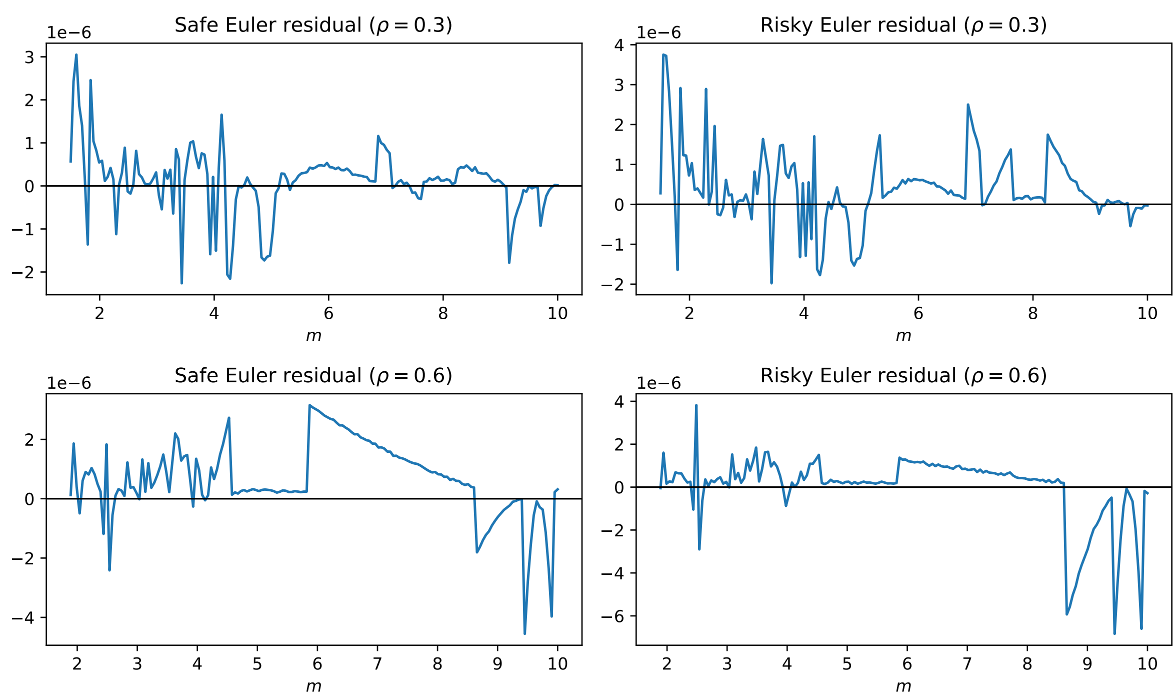


Figure 10: Euler residual diagnostics for the safe and risky asset, computed using the Epstein–Zin SDF (29) and masked to (approximate) interior points where $x \in (0, 1)$ and $x_s \in (0, 1)$.